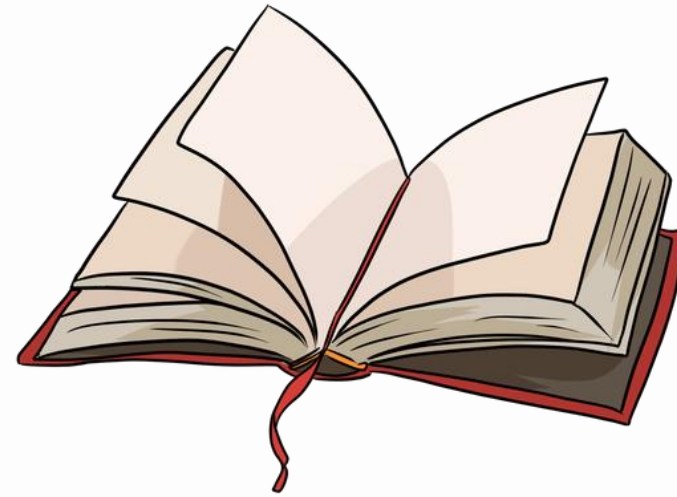




SCHOOL HELPER

SE GROUP 5

TABLE OF CONTENTS

01 What is school helper

02 About our project

03 Conclusion

04 Live demo

WHAT IS SCHOOL HELPER?

ABOUT OUR PROJECT

DOCUMENTS

PKM

Knowledge Manager

Documents

Knowledge Map

Summaries

Questions

Chat

7

docs

53

chunks

13

Q&A

General

New subject name

Create

Search your knowledge base...

Search

Documents

Upload PDF

Add URL

Add Text

Drag & drop PDF files here

or click to browse

PDF

1.Introduction to SE(new)

8 chunks 2026. 5. 6. 6 connections

Delete

PDF

2.Systems Engineering(new)

6 chunks 2026. 5. 6. 6 connections

Delete

PDF

3.Software Processes(new)

8 chunks 2026. 5. 6. 6 connections

Delete

PDF

4.Agile Software Development(new)

8 chunks 2026. 5. 6. 6 connections

Delete

PDF

5.Requirements Engineering(new)

10 chunks 2026. 5. 6. 6 connections

Delete

PDF

6.System Modeling(new)

6 chunks 2026. 5. 6. 6 connections

Delete

PDF

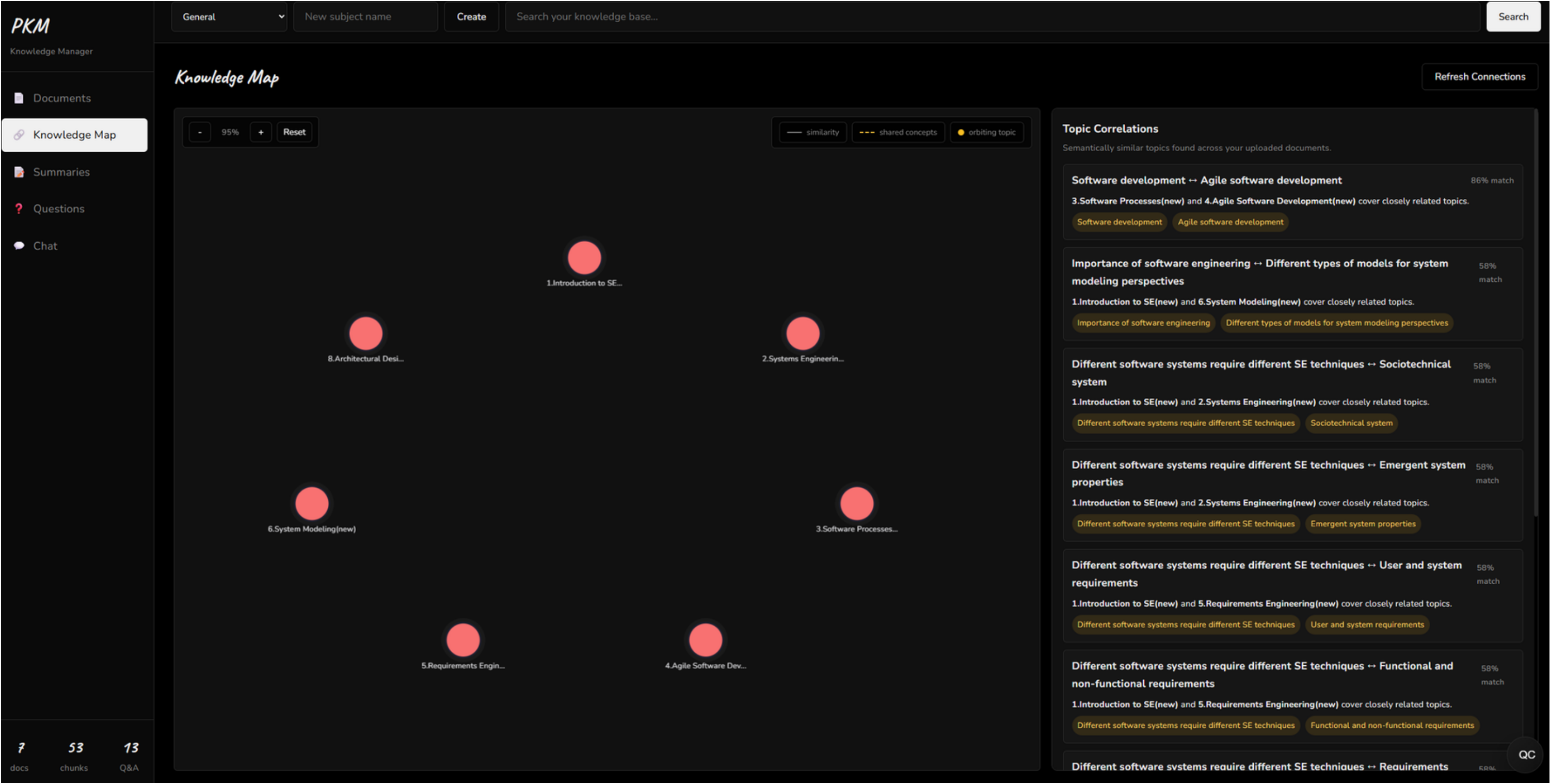
8.Architectural Design(new)

7 chunks 2026. 5. 15. 6 connections

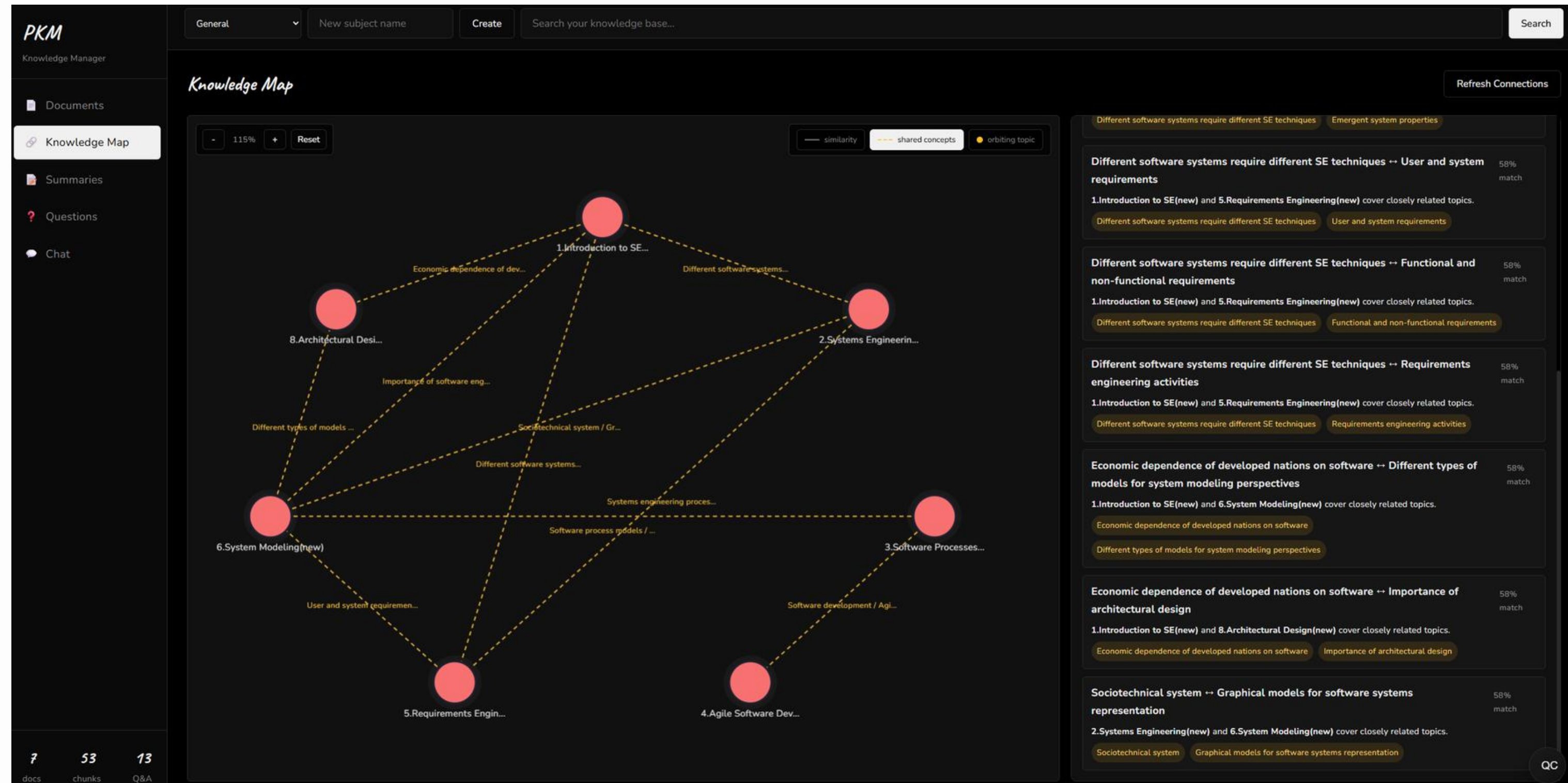
Delete

QC

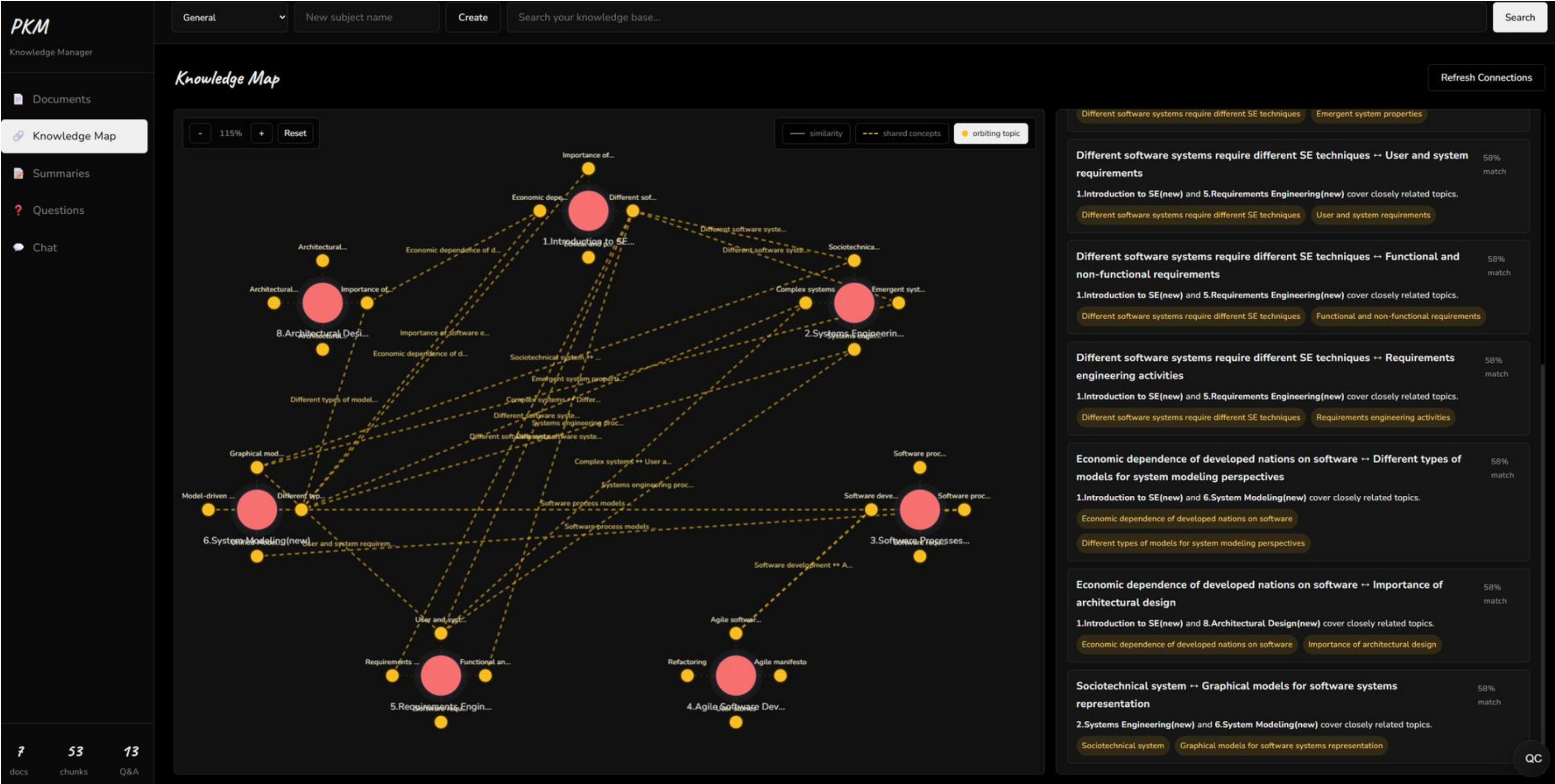
KNOWLEDGE MAP



KNOWLEDGE MAP



KNOWLEDGE MAP



SUMMARIES

PKM

Knowledge Manager

Documents

Knowledge Map

Summaries

Questions

Chat

General

New subject name

Create

Search your knowledge base...

Search

Summaries

PDF 1.Introduction to SE(new)

The document titled "Introduction to SE(new)" by Prof. Eunseok Lee provides a comprehensive overview of software engineering (SE), its significance, methodologies, ethical considerations, and various application types. The introduction emphasizes the critical role of software in modern economies, highlighting that software engineering is essential for developing reliable and cost-effective software systems. The document outlines the objectives of the course, which include understanding the importance of SE, recognizing different techniques for various software types, and addressing ethical issues relevant to software engineers.

Key concepts discussed include the distinction between generic and customized software products, the importance of software costs, and the various software engineering processes such as specification, development, validation, and evolution. The document also introduces the fundamental attributes of good software, which include maintainability, dependability, efficiency, and acceptability. It emphasizes that software engineering is not just about technical skills but also involves project management and adherence to ethical standards, as outlined in the ACM/IEEE Code of Ethics.

The document further explores the diversity of software engineering, noting that different types of applications—such as stand-alone systems, embedded control systems, and web-based applications—require tailored methodologies and tools. It also discusses the impact of the web on software engineering practices, emphasizing the importance of software reuse and incremental development in web-based systems. The latter sections of the document present case studies, including an insulin pump control system, a mental health care patient management system, and a wilderness weather station, illustrating practical applications of software engineering principles in real-world scenarios.

In conclusion, the document underscores the necessity for software engineers to uphold ethical standards and professional responsibilities while adapting to the evolving landscape of software development. It highlights the importance of continuous learning and ethical practice in ensuring that software engineering remains a respected and beneficial profession. The case studies serve to contextualize theoretical concepts, demonstrating their application in critical systems that impact health and safety.

▲

PDF 2.Systems Engineering(new)

▼

PDF 3.Software Processes(new)

▼

PDF 4.Agile Software Development(new)

▼

PDF 5.Requirements Engineering(new)

▼

PDF 6.System Modeling(new)

▼

PDF 8.Architectural Design(new)

▼

7docs

53chunks

13Q&A

QC

CHAT

PKM

Knowledge Manager

Documents

Knowledge Map

Summaries

Questions

Chat

General

New subject name

Create

Search your knowledge base...

Search

Chat with your Knowledge

what is the waterfall model?

high

The waterfall model is a software development process that involves a sequential design approach, where the project is divided into distinct stages. Each stage must be completed before the next one begins, making it inflexible in responding to changing customer requirements. This model is most suitable when the requirements are well-understood and changes during the design process are expected to be limited. It is primarily used for large systems engineering projects, where its plan-driven nature helps coordinate work across multiple sites.

The concept of the waterfall model appears in multiple documents, specifically in Document [1].

Document [1] provides a detailed explanation of the waterfall model, including its problems and appropriate use cases. It emphasizes the inflexibility of the model and its suitability for projects with stable requirements.

There are no contradictions or expansions on this concept in the other documents provided, as they focus on different aspects of software engineering and system modeling rather than specifically addressing the waterfall model.

Sources: 3.Software Processes(new), 6.System Modeling(new), 6.System Modeling(new), 6.System Modeling(new), 6.System Modeling(new), 1.Introduction to SE(new), 6.System Modeling(new), 6.System Modeling(new), 8.Architectural Design(new), 5.Requirements Engineering(new)

Also found in:

6.System Modeling(new).pdf

(example)

1.Introduction to SE(new).pdf

(example)

8.Architectural Design(new).pdf

(definition)

waterfall 모델이 뭐야?

high

워터폴 모델은 소프트웨어 개발 프로세스의 일종으로, 프로젝트를 여러 개의 단계로 나누어 순차적으로 진행하는 방식입니다. 각 단계는 이전 단계가 완료되어야만 시작할 수 있으며, 이로 인해 고객 요구 사항의 변화에 유연하게 대응하기 어렵습니다. 이 모델은 요구 사항이 잘 이해되고 변경이 제한적일 때 적합합니다. 주로 대규모 시스템 엔지니어링 프로젝트에서 사용되며, 계획 중심의 특성이 여러 사이트에서의 작업 조정을 돕습니다.

이 개념은 여러 문서에서 나타납니다. 문서 [1]에서는 워터폴 모델의 문제점과 적합한 사용 사례를 설명하고 있으며, 문서 [3]에서도 소프트웨어 공학의 다양한 개발 모델 중 하나로 언급되고 있습니다.

문서 [1]은 워터폴 모델의 장단점과 사용 조건을 구체적으로 설명하며, 문서 [3]은 소프트웨어 공학의 전반적인 개념을 다루면서 워터폴 모델을 포함한 다양한 개발 모델을 소개합니다. 따라서 문서 [1]이 워터폴 모델에 대한 구체적인 설명을 제공하고, 문서 [3]은 그 모델을 다른 개발 모델들과 함께 맥락에서 설명하는 방식으로 서로 관련되어 있습니다.

Ask a question...

Send

7docs

53chunks

15Q&A

QUICK CHAT

PKM

Knowledge Manager

Documents

Knowledge Map

Summaries

Questions

Chat

7 docs

53 chunks

16 Q&A

General

New subject name

Create

Search your knowledge base...

Search

Knowledge Map

Refresh Connections

- 115% + Reset

— similarity

--- shared concepts

● orbiting topic

Different software systems require different SE techniques

Emergent system properties

Different software systems require different SE techniques ↔ User and system requirements

58% match

1.Introduction to SE(new) and 5.Requirements Engineering(new) cover closely related topics.

Different software systems require different SE techniques

User and system requirements

Different software systems require different SE techniques ↔ Functional and non-functional requirements

58% match

1.Introduction to SE(new) and 5.Requirements Engineering(new) cover closely related topics.

Different software systems require different SE techniques

Functional and non-functional requirements

Different software systems require different SE techniques ↔ Requirements engineering activities

58% match

1.Introduction to SE(new) and 5.Requirements Engineering(new) cover closely related topics.

Different software systems require different SE techniques

Requirements engineering activities

Economic dependence of developed nations on software ↔ Different types of models for system modeling perspectives

58%

1.Introduction to SE(new) and 6.System Modeling(new) cover closely related topics.

Economic dependence of developed nations on software

Different types of models for system modeling perspectives

Economic dependence of developed nations on software ↔ Architectural design

58%

1.Introduction to SE(new) and 8.Architectural Design(new) cover closely related topics.

Economic dependence of developed nations on software

Architectural design

Sociotechnical system ↔ Graphical models for software representation

58%

2.Systems Engineering(new) and 6.System Modeling(new) cover closely related topics.

Sociotechnical system

Graphical models for software systems

Quick Chat

Same knowledge assistant

requirement specification이 뭐야?

high

요구 사항 명세(requirement specification)는 소프트웨어 개발 과정에서 시스템의 사용자 요구 사항과 시스템 요구 사항을 문서화하는 프로세스를 의미합니다. 이 문서는 시스템이 제공해야 하는 기능과 서비스

Ask a question...

Send

QUICK CHAT

PKM Knowledge Manager

General ▾ New subject name Create Search your knowledge base... Search

Knowledge Map

Documents
Summaries
Questions
Chat

Knowledge Map

115% Reset

— similarity --- shared concepts ● orbiting topic

8.Architectural Desi...
1.Introduction to SE...
2.Systems Engineerin...
3.Software Processes...
4.Agile Software Dev...
5.Requirements Engin...
6.System Modeling(new)
Economic dependence of dev...
Different software systems...
Importance of software eng...
Different types of models ...
Sociotechnical system / Gr...
Different software systems...
Systems engineering proces...
Software process models / ...
User and system requiremen...
Software development / Agi...

Topic Correlations

Semantically similar topics found across your uploaded documents.

Software development ↔ Agile software development 86% match
3.Software Processes(new) and 4.Agile Software Development(new) cover closely related topics.
Software development Agile software development

Importance of software engineering ↔ Different types of models for system modeling perspectives 58% match
1.Introduction to SE(new) and 6.System Modeling(new) cover closely related topics.
Importance of software engineering Different types of models for system modeling perspectives

Different software systems require different SE techniques ↔ Sociotechnical system 58% match
1.Introduction to SE(new) and 2.Systems Engineering(new) cover closely related topics.
Different software systems require different SE techniques Sociotechnical system

Different software systems require different SE techniques ↔ Emergent system properties 58% match
1.Introduction to SE(new) and 2.Systems Engineering(new) cover closely related topics.
Different software systems require different SE techniques Emergent system properties

Different software systems require different SE techniques ↔ User and system requirements 58% match
1.Introduction to SE(new) and 5.Requirements Engineering(new) cover closely related topics.
Different software systems require different SE techniques User and system requirements

Different software systems require different SE techniques ↔ Functional and non-functional requirements 58% match
1.Introduction to SE(new) and 5.Requirements Engineering(new) cover closely related topics.
Different software systems require different SE techniques Functional and non-functional requirements

Different software systems require different SE techniques ↔ Requirements 58% match
1.Introduction to SE(new) and 5.Requirements Engineering(new) cover closely related topics.
Different software systems require different SE techniques Requirements

7 docs 53 chunks 16 Q&A

QC

QUESTIONS

PKM

Knowledge Manager

Documents

Knowledge Map

Summaries

Questions

Chat

7docs

53chunks

16Q&A

General

New subject name

Create

Search your knowledge base...

Search

Study Questions

Reload Questions

By PDF

Select a document to study its question set.

1.Introduction to SE(new)

6 questions · Importance of software engineering, Different software systems require different SE techniques

2.Systems Engineering(new)

6 questions · Sociotechnical system, Emergent system properties

3.Software Processes(new)

6 questions · Software process, Software process models

4.Agile Software Development(new)

0 questions · Agile software development, Agile manifesto

5.Requirements Engineering(new)

6 questions · User and system requirements, Functional and non-functional requirements

6.System Modeling(new)

6 questions · Graphical models for software systems representation, Different types of models for system modeling perspectives

8.Architectural Design(new)

0 questions · Architectural design, Importance of architectural design

1.Introduction to SE(new) · 6 questions

Short Answer

Multiple Choice

Short Answer

Generate Questions

Adaptive Next

attributes of good software

1.Introduction to SE(new)

What are the key attributes of good software, and why is each attribute important?

Write your answer here...

Submit Answer

Answer History

Short Answer

Score: 90%

Explain how maintainability impacts the overall cost of software over its lifecycle.

Your answer:
Maintainability has a major impact on the overall cost of software because most costs occur after deployment during maintenance and updates. Software with high maintainability is easier to modify, debug, and extend, reducing the time and effort required for future changes. This lowers maintenance costs, speeds up feature development, and helps the software adapt to changing requirements throughout its lifecycle.

Feedback:
The student's answer effectively explains the impact of maintainability on overall software cost by highlighting that most costs occur post-deployment during maintenance and updates. It correctly notes that high maintainability reduces time and effort for modifications, debugging, and extensions, which lowers maintenance costs and facilitates adaptation to changing requirements. However, the response could be improved by explicitly mentioning that poor maintainability can lead to maintenance costs exceeding initial development costs, especially for long-lifetime systems, as stated in the expected answer.

2026-06-08T22:41:13.919635

Short Answer

Score: 85%

QC

QUESTIONS

PKM

Knowledge Manager

Documents

Knowledge Map

Summaries

Questions

Chat

7

53

16

docs

chunks

Q&A

General

New subject name

Create

Search your knowledge base...

Search

Study Questions

Reload Questions

By PDF

Select a document to study its question set.

1.Introduction to SE(new)
6 questions · Importance of software engineering, Different software systems require different SE techniques

2.Systems Engineering(new)
6 questions · Sociotechnical system, Emergent system properties

3.Software Processes(new)
6 questions · Software process, Software process models

4.Agile Software Development(new)
0 questions · Agile software development, Agile manifesto

5.Requirements Engineering(new)
6 questions · User and system requirements, Functional and non-functional requirements

6.System Modeling(new)
6 questions · Graphical models for software systems representation, Different types of models for system modelling perspectives

8.Architectural Design(new)
0 questions · Architectural design, Importance of architectural design

1.Introduction to SE(new) · 6 questions

Multiple Choice

Generate Questions

Adaptive Next

attributes of good software

EASY

1.Introduction to SE(new)

Which of the following is NOT considered an attribute of good software?

Maintainability

Dependability

Complexity

Efficiency

Answer History

Short Answer

Score: 90%

Explain how maintainability impacts the overall cost of software over its lifecycle.

Your answer:
Maintainability has a major impact on the overall cost of software because most costs occur after deployment during maintenance and updates. Software with high maintainability is easier to modify, debug, and extend, reducing the time and effort required for future changes. This lowers maintenance costs, speeds up feature development, and helps the software adapt to changing requirements throughout its lifecycle.

Feedback:
The student's answer effectively explains the impact of maintainability on overall software cost by highlighting that most costs occur post-deployment during maintenance and updates. It correctly notes that high maintainability reduces time and effort for modifications, debugging, and extensions, which lowers maintenance costs and facilitates adaptation to changing requirements. However, the response could be improved by explicitly mentioning that poor maintainability can lead to maintenance costs exceeding initial development costs, especially for long-lifetime systems, as stated in the expected answer.

2026-06-08T22:41:13.919635

Short Answer

Score: 85%

...

QC

QUESTIONS

PKM

Knowledge Manager

Documents

Knowledge Map

Summaries

?

Questions

Chat

7

docs

53

chunks

16

Q&A

General

New subject name

Create

Search your knowledge base...

Search

Study Questions

Reload Questions

By PDF

Select a document to study its question set.

1.Introduction to SE(new)
6 questions · Importance of software engineering, Different software systems require different SE techniques

2.Systems Engineering(new)
6 questions · Sociotechnical system, Emergent system properties

3.Software Processes(new)
6 questions · Software process, Software process models

4.Agile Software Development(new)
0 questions · Agile software development, Agile manifesto

5.Requirements Engineering(new)
6 questions · User and system requirements, Functional and non-functional requirements

6.System Modeling(new)
6 questions · Graphical models for software systems representation, Different types of models for system modeling perspectives

8.Architectural Design(new)
0 questions · Architectural design, Importance of architectural design

1.Introduction to SE(new) · 6 questions

Multiple Choice

Generate Questions

Adaptive Next

attributes of good software

EASY

1.Introduction to SE(new)

Which of the following is NOT considered an attribute of good software?

Maintainability

Dependability

Complexity

Efficiency

Correct

Complexity is not an attribute of good software; rather, good software should aim for simplicity and ease of use.

Topic mastery: 25%

Next Adaptive Question

Answer History

Short Answer

Score: 90%

Explain how maintainability impacts the overall cost of software over its lifecycle.

Your answer:

Maintainability has a major impact on the overall cost of software because most costs occur after deployment during maintenance and updates. Software with high maintainability is easier to modify, debug, and extend, reducing the time and effort required for future changes. This lowers maintenance costs, speeds up feature development, and helps the software adapt to changing requirements throughout its lifecycle.

Feedback:

The student's answer effectively explains the impact of maintainability on overall software cost by highlighting that most costs occur post-deployment during maintenance and updates. It correctly notes that high maintainability reduces time and effort for modifications, debugging, and extensions, which lowers maintenance costs and facilitates adaptation to changing requirements. However, the response could be improved by explicitly mentioning that poor maintainability can lead to maintenance costs exceeding initial development costs, especially for long-lifetime systems, as stated in the expected answer.

QC

QUESTIONS

PKM

Knowledge Manager

Documents

Knowledge Map

Summaries

Questions

Chat

7docs

53chunks

16Q&A

General

New subject name

Create

Search your knowledge base...

Search

Study Questions

Reload Questions

By PDF

Select a document to study its question set.

1.Introduction to SE(new)
6 questions · Importance of software engineering, Different software systems require different SE techniques

2.Systems Engineering(new)
6 questions · Sociotechnical system, Emergent system properties

3.Software Processes(new)
6 questions · Software process, Software process models

4.Agile Software Development(new)
0 questions · Agile software development, Agile manifesto

5.Requirements Engineering(new)
6 questions · User and system requirements, Functional and non-functional requirements

6.System Modeling(new)
6 questions · Graphical models for software systems representation, Different types of models for system modeling perspectives

8.Architectural Design(new)
0 questions · Architectural design, Importance of architectural design

1.Introduction to SE(new) · 6 questions

Short Answer

Generate Questions

Adaptive Next

attributes of good software

MEDIUM

1.Introduction to SE(new)

What are the key attributes of good software, and why is each attribute important?

Good software should be maintainable, dependable, secure, efficient, and acceptable to its users. Maintainability is important because software needs to be updated, fixed, and improved over time. Dependability and security ensure that the system works reliably and protects data from errors and attacks. Efficiency helps the software provide good performance while using resources effectively. Acceptability is important because users must be able to use the software easily and find it suitable for their needs and working environment.

Submit Answer

Answer History

Short Answer

Score: 90%

Explain how maintainability impacts the overall cost of software over its lifecycle.

Your answer:
Maintainability has a major impact on the overall cost of software because most costs occur after deployment during maintenance and updates. Software with high maintainability is easier to modify, debug, and extend, reducing the time and effort required for future changes. This lowers maintenance costs, speeds up feature development, and helps the software adapt to changing requirements throughout its lifecycle.

Feedback:
The student's answer effectively explains the impact of maintainability on overall software cost by highlighting that most costs occur post-deployment during maintenance and updates. It correctly notes that high maintainability reduces time and effort for modifications, debugging, and extensions, which lowers maintenance costs and facilitates adaptation to changing requirements. However, the response could be improved by explicitly mentioning that poor maintainability can lead to maintenance costs exceeding initial development costs, especially for long-lifetime systems, as stated in the expected answer.

2026-06-08T22:41:13.919635

Multiple Choice

Correct

QC

QUESTIONS

PKM

Knowledge Manager

Documents

Knowledge Map

Summaries

Questions

Chat

7docs

53chunks

16Q&A

General

New subject name

Create

Search your knowledge base...

Search

Study Questions

Reload Questions

By PDF

Select a document to study its question set.

1.Introduction to SE(new)
6 questions · Importance of software engineering, Different software systems require different SE techniques

2.Systems Engineering(new)
6 questions · Sociotechnical system, Emergent system properties

3.Software Processes(new)
6 questions · Software process, Software process models

4.Agile Software Development(new)
0 questions · Agile software development, Agile manifesto

5.Requirements Engineering(new)
6 questions · User and system requirements, Functional and non-functional requirements

6.System Modeling(new)
6 questions · Graphical models for software systems representation, Different types of models for system modeling perspectives

8.Architectural Design(new)
0 questions · Architectural design, Importance of architectural design

1.Introduction to SE(new) · 6 questions

Short Answer

Generate Questions

Adaptive Next

attributes of good software

MEDIUM

1.Introduction to SE(new)

What are the key attributes of good software, and why is each attribute important?

Write your answer here...

Submit Answer

Score: 90%

Topic mastery: 90%

AI Feedback:

The student correctly identifies most of the key attributes: maintainability, dependability, efficiency, and acceptability, and provides clear explanations for each. The inclusion of 'security' is a good addition, though it was not explicitly requested in the question. The explanation of dependability is enhanced by mentioning security, which is relevant but slightly diverges from the expected answer. The term 'acceptable' is used instead of 'acceptability,' but the meaning is clear. Overall, the response is thorough, well-articulated, and demonstrates a good understanding of the importance of each attribute. To improve, the student could align more closely with the expected attributes and avoid introducing extra concepts unless asked.

Sample Answer:

The key attributes of good software include maintainability, dependability, efficiency, and acceptability. Maintainability ensures that software can be easily updated and modified, which is crucial for long-term usability. Dependability means the software is reliable and performs consistently, which is essential for user trust. Efficiency relates to how well the software utilizes resources, impacting performance and cost-effectiveness. Acceptability refers to how well the software meets user needs and expectations, which is vital for user satisfaction.

Next Adaptive Question

Answer History

Short Answer

QC

QUESTIONS

PKM

Knowledge Manager

Documents

Knowledge Map

Summaries

Questions

Chat

7docs

53chunks

16Q&A

General

New subject name

Create

Search your knowledge base...

Search

Study Questions

Reload Questions

By PDF

Select a document to study its question set.

1.Introduction to SE(new)
6 questions · Importance of software engineering, Different software systems require different SE techniques

2.Systems Engineering(new)
6 questions · Sociotechnical system, Emergent system properties

3.Software Processes(new)
6 questions · Software process, Software process models

4.Agile Software Development(new)
0 questions · Agile software development, Agile manifesto

5.Requirements Engineering(new)
6 questions · User and system requirements, Functional and non-functional requirements

6.System Modeling(new)
6 questions · Graphical models for software systems representation, Different types of models for system modeling perspectives

8.Architectural Design(new)
0 questions · Architectural design, Importance of architectural design

Answer History

Short Answer

Score: 90%

Explain how maintainability impacts the overall cost of software over its lifecycle.

Your answer:
Maintainability has a major impact on the overall cost of software because most costs occur after deployment during maintenance and updates. Software with high maintainability is easier to modify, debug, and extend, reducing the time and effort required for future changes. This lowers maintenance costs, speeds up feature development, and helps the software adapt to changing requirements throughout its lifecycle.

Feedback:
The student's answer effectively explains the impact of maintainability on overall software cost by highlighting that most costs occur post-deployment during maintenance and updates. It correctly notes that high maintainability reduces time and effort for modifications, debugging, and extensions, which lowers maintenance costs and facilitates adaptation to changing requirements. However, the response could be improved by explicitly mentioning that poor maintainability can lead to maintenance costs exceeding initial development costs, especially for long-lifetime systems, as stated in the expected answer.

2026-06-08T22:41:13.919635

Multiple Choice

Correct

Which of the following is NOT considered an attribute of good software?

Your answer:
Complexity

Correct answer:
Complexity

Explanation:
Complexity is not an attribute of good software; rather, good software should aim for simplicity and ease of use.

2026-06-08T14:23:56.511729+00:00

Short Answer

Score: 85%

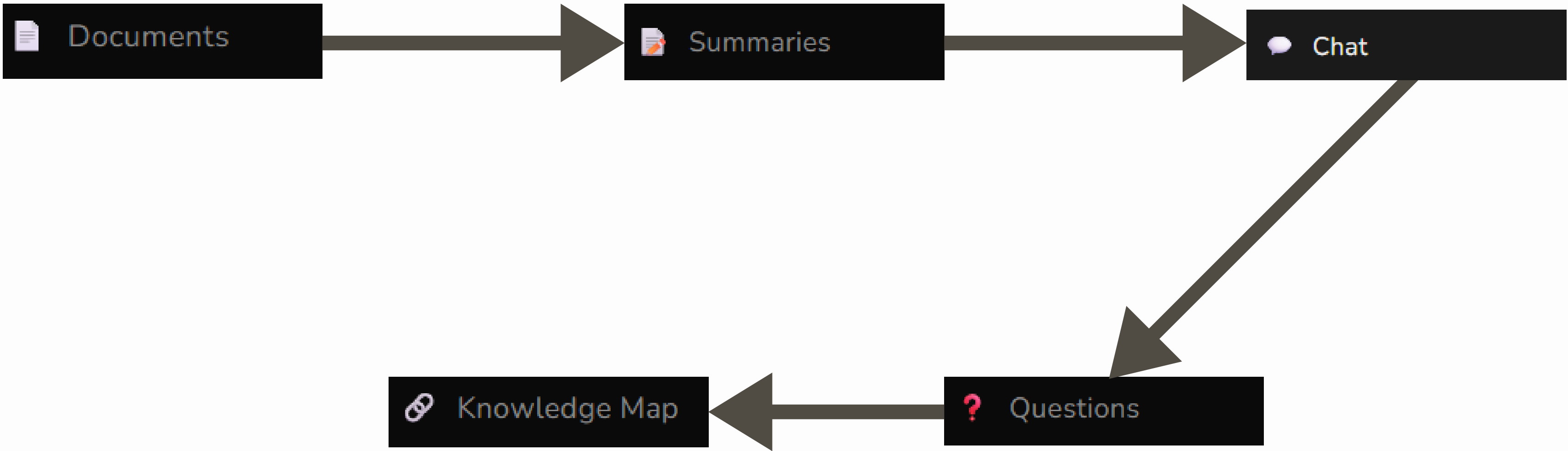
What are the key attributes of a good software process, and why are they important?

Your answer:
Understandability The process should be easy for developers and stakeholders to understand. This helps teams follow the process consistently and reduces confusion during

2026-06-08T14:23:56.511729+00:00

QC

CONCLUSION



LIVE DEMO

THANK YOU